```
import pandas as pd
```

```
df=pd.read_csv('/content/Salary_dataset.csv')
```

```
df.head()
```

	Unnamed: 0	YearsExperience	Salary
0	0	1.2	39344.0
1	1	1.4	46206.0
2	2	1.6	37732.0
3	3	2.1	43526.0
4	4	2.3	39892.0

```
df.tail()
```

	Unnamed: 0	YearsExperience	Salary
25	25	9.1	105583.0
26	26	9.6	116970.0
27	27	9.7	112636.0
28	28	10.4	122392.0
29	29	10.6	121873.0

```
x=df['YearsExperience'].values
y=df['Salary'].values
```

```
x
```

```
array([ 1.2,  1.4,  1.6,  2.1,  2.3,  3. ,  3.1,  3.3,  3.3,  3.8,  4. ,
        4.1,  4.1,  4.2,  4.6,  5. ,  5.2,  5.4,  6. ,  6.1,  6.9,  7.2,
        8. ,  8.3,  8.8,  9.1,  9.6,  9.7, 10.4, 10.6])
```

```
y
```

```
array([ 39344.,  46206.,  37732.,  43526.,  39892.,  56643.,  60151.,
        54446.,  64446.,  57190.,  63219.,  55795.,  56958.,  57082.,
        61112.,  67939.,  66030.,  83089.,  81364.,  93941.,  91739.,
        98274., 101303., 113813., 109432., 105583., 116970., 112636.,
       122392., 121873.])
```

```
x=df.iloc[:, 1].values
```

```
x
```

```
array([ 1.2,  1.4,  1.6,  2.1,  2.3,  3. ,  3.1,  3.3,  3.3,  3.8,  4. ,
        4.1,  4.1,  4.2,  4.6,  5. ,  5.2,  5.4,  6. ,  6.1,  6.9,  7.2,
        8. ,  8.3,  8.8,  9.1,  9.6,  9.7, 10.4, 10.6])
```

```
y=df.iloc[:, 2].values
```

```
y
```

```
array([ 39344.,  46206.,  37732.,  43526.,  39892.,  56643.,  60151.,
        54446.,  64446.,  57190.,  63219.,  55795.,  56958.,  57082.,
        61112.,  67939.,  66030.,  83089.,  81364.,  93941.,  91739.,
        98274., 101303., 113813., 109432., 105583., 116970., 112636.,
       122392., 121873.])
```

```
import numpy as np
```

```
X=np.array(x)
Y=np.array(y)
```

```
X=X.reshape(-1,1)
X
```

```
array([[ 1.2],
       [ 1.4],
       [ 1.6],
       [ 2.1],
       [ 2.3],
       [ 3. ],
       [ 3.1],
       [ 3.3],
       [ 3.3],
       [ 3.8],
       [ 4. ],
       [ 4.1],
       [ 4.1],
       [ 4.2],
       [ 4.6],
       [ 5. ],
       [ 5.2],
       [ 5.4],
       [ 6. ],
       [ 6.1],
       [ 6.9],
       [ 7.2],
       [ 8. ],
       [ 8.3],
       [ 8.8],
       [ 9.1],
       [ 9.6],
       [ 9.7],
       [10.4],
       [10.6]])
```

```
Y=Y.reshape(-1,1)
Y
```

```
array([[ 39344.],
       [ 46206.],
       [ 37732.],
       [ 43526.],
       [ 39892.],
       [ 56643.],
       [ 60151.],
       [ 54446.],
       [ 64446.],
       [ 57190.],
       [ 63219.],
       [ 55795.],
       [ 56958.],
       [ 57082.],
       [ 61112.],
       [ 67939.],
       [ 66030.],
       [ 83089.],
       [ 81364.],
       [ 93941.],
       [ 91739.],
       [ 98274.],
       [101303.],
       [113813.],
       [109432.],
       [105583.],
       [116970.],
       [112636.],
       [122392.],
       [121873.]])
```

```
X_min=X.min()
X_max=X.max()
X_norm=(X-X_min)/(X_max-X_min)
X_norm
```

```
array([[0.        ],
       [0.0212766 ],
       [0.04255319],
       [0.09574468],
       [0.11702128],
       [0.19148936],
       [0.20212766],
       [0.22340426],
       [0.22340426],
       [0.27659574],
       [0.29787234],
       [0.30851064],
       [0.30851064],
       [0.31914894],
       [0.36170213],
       [0.40425532],
       [0.42553191],
       [0.44680851],
       [0.5106383 ],
```

```
       [0.5212766 ],
       [0.60638298],
       [0.63829787],
       [0.72340426],
       [0.75531915],
       [0.80851064],
       [0.84042553],
       [0.89361702],
       [0.90425532],
       [0.9787234 ],
       [1.        ]])
```

```
m=np.random.randn()
c=np.random.randn()
```

```
m
```

```
-0.7337434621873196
```

```
c
```

```
0.8117597632097044
```

```
def predict(X, m, c):
  y_pred=m*X+c
  return y_pred
```

```
y_pred=predict(X, m, c)
```

```
y_pred
```

```
array([[-0.06873239],
       [-0.21548108],
       [-0.36222978],
       [-0.72910151],
       [-0.8758502 ],
       [-1.38947062],
       [-1.46284497],
       [-1.60959366],
       [-1.60959366],
       [-1.97646539],
       [-2.12321409],
       [-2.19658843],
       [-2.19658843],
       [-2.26996278],
       [-2.56346016],
       [-2.85695755],
       [-3.00370624],
       [-3.15045493],
       [-3.59070101],
       [-3.66407536],
       [-4.25107013],
       [-4.47119316],
       [-5.05818793],
       [-5.27831097],
       [-5.6451827 ],
       [-5.86530574],
       [-6.23217747],
       [-6.30555182],
       [-6.81917224],
       [-6.96592094]])
```

```
print("first 5 predicted values:")
print(y_pred[:5])
```

```
first 5 predicted values:
[[-0.06873239]
 [-0.21548108]
 [-0.36222978]
 [-0.72910151]
 [-0.8758502 ]]
```

```
def compute_cost(y, y_pred):
  n=len(y)
  cost=np.sum((y-y_pred)**2)/n
  return cost
```

```
cost = compute_cost(Y, y_pred)
print("initial cost (MSE):", cost)
```

```
initial cost (MSE): 6503695635.568552
```

```
m=0.0
c=0.0
y_pred=predict(X, m, c)
cost=compute_cost(Y, y_pred)
print("cost for m=",m, "c=", c, "is", cost)
```

```
cost for m= 0.0 c= 0.0 is 6503107277.733334
```

```
learning_rate=0.001
epochs=2000
```

```
learning_rate
```

```
0.001
```

```
epochs
```

```
2000
```

```
n=len(y)
```

```
cost_history=[]
```

```
for epoch in range(epochs):
  y_pred=predict(X, m, c)
  dm=(1/n)*np.sum((y_pred-y) * X_norm)
  dc=(1/n)*np.sum((y_pred-y))
  m=m-learning_rate*dm
  c=c-learning_rate*dc
  cost=compute_cost(Y, y_pred)
  cost_history.append(cost)
  if epoch%100==0:
    print(f"epoch {epoch}: cost={cost}")
```

```
epoch 0: cost=726499261.7333314
epoch 100: cost=726499261.7333316
epoch 200: cost=726499261.7333316
epoch 300: cost=726499261.7333316
epoch 400: cost=726499261.7333316
epoch 500: cost=726499261.7333316
epoch 600: cost=726499261.7333316
epoch 700: cost=726499261.7333316
epoch 800: cost=726499261.7333316
epoch 900: cost=726499261.7333316
epoch 1000: cost=726499261.7333316
epoch 1100: cost=726499261.7333316
epoch 1200: cost=726499261.7333316
epoch 1300: cost=726499261.7333316
epoch 1400: cost=726499261.7333316
epoch 1500: cost=726499261.7333316
epoch 1600: cost=726499261.7333316
epoch 1700: cost=726499261.7333316
epoch 1800: cost=726499261.7333316
epoch 1900: cost=726499261.7333316
```

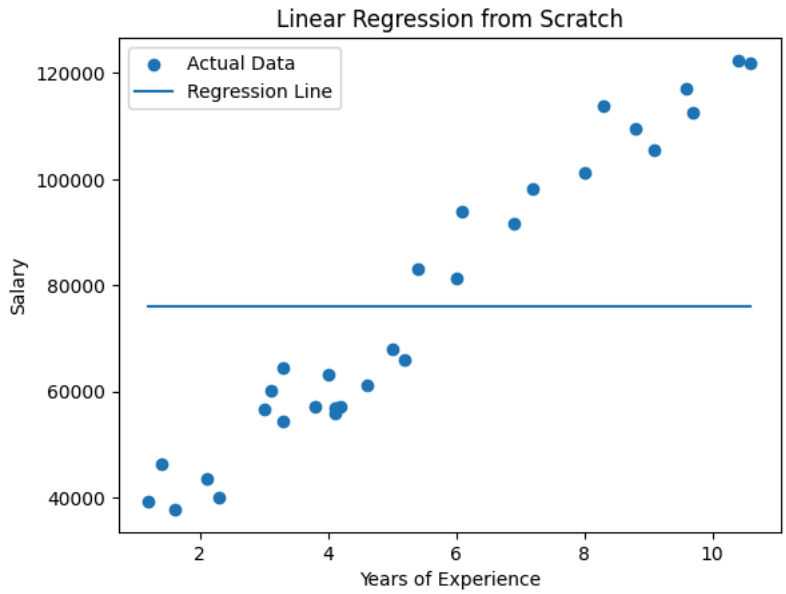
```
final_slope=m
final_intercept=c
print("final model parameters:")
print("slope (m):", final_slope)
print("intercept (c):", final_intercept)
```

```
final model parameters:
slope (m): 1.1899998691204716e-11
intercept (c): 76003.99999999991
```

```
def predict_salary(years_experience, m, c, X_min, X_max):
  # Normalize the single input years_experience
  normalized_years_experience = (years_experience - X_min) / (X_max - X_min)
  # Predict using the normalized value and the learned parameters
  salary_pred = m * normalized_years_experience + c
  return salary_pred
```

```
years=5
predicted_salary=predict_salary(years, final_slope, final_intercept, X_min, X_max)
print("predicted salary for", years, "years of experience:", predicted_salary)
```

```
predicted salary for 5 years of experience: 76003.99999999991
```

```
import matplotlib.pyplot as plt
```

```
y_pred_final=predict(X, final_slope, final_intercept)
plt.scatter(X, y, label="Actual Data")
plt.plot(X, y_pred_final, label="Regression Line")
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Linear Regression from Scratch")
plt.legend()
plt.show()
```